

run potential of stocks to outperform bonds and provide positive real returns, so investors are generally advised to take on as much risk as they can tolerate in order to minimize the probability of failure.

Probability advocates generally see little value in income annuities. Income annuities have no upside potential, a cost these advocates view as too high regardless of the safety the income annuity provides. Especially with today's low interest rates, building a lifetime floor in such a way can be seen as unnecessarily expensive. Income annuities may protect a person from destitution, but, probability-based advocates argue, they could also lock out the ability to enjoy the higher quality of life people desire for their retirement.

Most retirees will not have saved enough to safely immunize their entire lifestyle spending goals through only bond ladders and income annuities while still maintaining sufficient remaining wealth to create a liquid contingency fund for unexpected expenses. If a retiree's desired withdrawal rate is above what can be generated with the bond yield curve, a bond portfolio will not be able to meet his or her spending goals. Bonds would actually become a drag on the portfolio, as they offer no chance at the types of returns needed to fund the retiree's desired lifestyle. It is the same if the spending goal exceeds what can be obtained with an income annuity. Equity exposure moves retirees away from the guarantee that their plan will work, but it might provide the only opportunity for them to meet all their aspirations. This aspect of probability maximization through a diversified portfolio is why we refer to this school of thought as "probability-based."

*What is the safe withdrawal rate from a diversified portfolio of volatile assets?*

Users of safe withdrawal rates generally treat 4 percent as a reasonably safe worst-case sustainable withdrawal rate for a thirty-year retirement period, although they acknowledge that a new worst-case scenario could force that number lower. Bengen now speaks regularly about 4.5 percent as the safe withdrawal rate, a result of also including small-capitalization stocks in the portfolio mix. He is confident that US history provides a good guide about